

A basic-hypergeometric form of the partition-matrix inversion series

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The original problem

A *partition matrix* on $[n]$ is an upper-triangular square matrix of subsets of $[n] = \{1, \dots, n\}$ with the following three properties: every row and column has a nonempty cell; the nonempty cells partition $[n]$; and if $i < j$, then the column containing i is not to the right of the column containing j . Write $\text{row}(i)$ and $\text{col}(i)$ for the cell coordinates of i . Its inversion number is

$$\text{inv}(P) = \#\{(i, j) : i > j, \text{col}(i) = \text{col}(j), \text{row}(i) < \text{row}(j)\}.$$

For

$$S_n(q) = \sum_{P \in \text{PM}_n} q^{\text{inv}(P)},$$

Chern–Fu, Question 5.1 [2], asks for a closed expression for

$$\mathcal{F}(q, t) = \sum_{n \geq 1} S_n(q) t^n. \tag{P}$$

We first give a complete sum-product solution. We then put its column series in standard basic-hypergeometric form and extend the argument to arbitrary weights for cell cardinalities.

1 The complete sum-product solution

For an ordered alphabet $[k]$, let

$$H_{m,k}(q) = \sum_{w \in [k]^m} q^{\text{inv}(w)}, \quad R_k(q, t) = \sum_{m \geq 0} H_{m,k}(q) t^m.$$

If letter i occurs a_i times, the inversion enumerator of all rearrangements is the Gaussian multinomial. Thus

$$H_{m,k}(q) = \sum_{a_1 + \dots + a_k = m} \left[\begin{matrix} m \\ a_1, \dots, a_k \end{matrix} \right]_q.$$

These polynomials are the generalized Galois numbers; their word interpretation and multivariate Rogers–Szegő generating function are classical [3, 4].

Fix a partition matrix of dimension d . Its labels in column j form a nonempty consecutive interval. Read them in increasing order and record their row indices. This gives a nonempty word

$$w_j \in [j]^+.$$

Every row is nonempty precisely when every letter in $[d]$ occurs in at least one of $w_1, \dots, w_d$. Conversely, any such word sequence reconstructs a unique partition matrix: split $[n]$ into

consecutive intervals of lengths $|w_1|, \dots, |w_d|$, and place each label in the row specified by its letter. Moreover,

$$\text{inv}(P) = \sum_{j=1}^d \text{inv}(w_j),$$

because a matrix inversion is exactly a word inversion inside one column.

Put $W_k = R_k - 1$, the nonempty-word series, and $W_0 = 0$. Inclusion–exclusion over absent rows gives the dimension- d series

$$F_d(q, t) = \sum_{A \subseteq [d]} (-1)^{|A|} \prod_{j=1}^d W_{j-|A \cap [j]|}(q, t).$$

Indeed, after rows in A are forbidden, column j retains an ordered alphabet of size $j - |A \cap [j]|$.

The alternating sum has a simple path interpretation. Set $k_j = j - |A \cap [j]|$. The path k_j rises when $j \notin A$ and stays level when $j \in A$. A rise into level k contributes W_k ; a stay contributes $-W_k$. A nonzero path begins with a rise because $W_0 = 0$. After summing over all dimensions, a path reaching level d has one rise into each attained level, followed by an arbitrary number of stays. Their total weight at level k is

$$\sum_{s \geq 0} W_k (-W_k)^s = \frac{W_k}{1 + W_k} = 1 - R_k^{-1}.$$

Multiplying over the levels proves the desired sum-product from first principles.

Theorem 1 (Answer to Question 5.1). *In $\mathbb{Z}[q][[t]]$,*

$$\mathcal{F}(q, t) = \sum_{d \geq 1} \prod_{k=1}^d (1 - R_k(q, t)^{-1}).$$

Proof. The column bijection, inclusion–exclusion, and path summation show that the coefficient of t^n is exactly the inversion enumerator of partition matrices on $[n]$. Each factor in the displayed product has positive t -adic order, so only $d \leq n$ contributes to $[t^n]$; the infinite sum is therefore a legitimate formal series. $\square$

The index d in the final sum counts attained path levels; after inclusion–exclusion it is no longer the dimension of the original matrix. This also explains why no separate row statistic appears in the product.

2 Putting the column series in standard form

Write

$$(a; q)_r = \prod_{j=0}^{r-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j \geq 0} (1 - aq^j).$$

The useful transform here is the normalized coefficient transform

$$\mathcal{B}_q \left(\sum_{m \geq 0} f_m t^m \right) = \sum_{m \geq 0} \frac{f_m}{(q; q)_m} z^m.$$

The Gaussian-multinomial expression makes its effect transparent:

$$\mathcal{B}_q R_k = \left(\sum_{a \geq 0} \frac{z^a}{(q; q)_a} \right)^k = \frac{1}{(z; q)_\infty^k}.$$

The last equality is Euler's q -binomial theorem.

We will also need an explicit inverse. For $f(z) = \sum b_m z^m$, define

$$\mathcal{L}_q f(t) = (q; q)_\infty \sum_{r \geq 0} \frac{q^r}{(q; q)_r} f(q^r t).$$

Lemma 2. *The two transforms satisfy $\mathcal{L}_q \mathcal{B}_q f = f$.*

Proof. The coefficient of t^m in $\mathcal{L}_q f$ is

$$b_m(q; q)_\infty \sum_{r \geq 0} \frac{q^{r(m+1)}}{(q; q)_r} = b_m \frac{(q; q)_\infty}{(q^{m+1}; q)_\infty} = b_m(q; q)_m,$$

again by Euler's identity. This exactly reverses coefficient normalization. $\square$

The calculation is coefficientwise in $\mathbb{Z}[[q]][[t]]$; when $|q| < 1$, it also has its usual analytic meaning near $t = 0$. Applying Lemma 2 to the product formula gives

$$\begin{aligned} R_k(q, t) &= (q; q)_\infty \sum_{r \geq 0} \frac{q^r}{(q; q)_r (q^r t; q)_\infty^k} \\ &= \frac{(q; q)_\infty}{(t; q)_\infty^k} \sum_{r \geq 0} \frac{(t; q)_r^k}{(q; q)_r} q^r. \end{aligned}$$

The last sum is a standard basic hypergeometric series: its r th summand is $(t; q)_r^k q^r / (q; q)_r$. Thus no new special function is needed.

Theorem 3 (The word series). *For every $k \geq 1$,*

$$R_k(q, t) = \frac{(q; q)_\infty}{(t; q)_\infty^k} {}_k\phi_{k-1} \left(\begin{matrix} t, \dots, t \\ 0, \dots, 0 \end{matrix}; q, q \right).$$

For $k = 1$ the formula reduces to $R_1 = (1 - t)^{-1}$; the next two cases use ${}_2\phi_1$ and ${}_3\phi_2$. Substitution in Theorem 1 gives the requested standard-form expression.

The analytic assertion concerns each fixed column series R_k . We make no assertion of analytic convergence for the outer sum over d .

3 The natural cell-size refinement

The content of a column word is exactly the list of cell cardinalities in that column. This gives a refinement without changing the proof. Let $\mathbf{u} = (u_1, u_2, \dots)$, put $u_0 = 1$, and give every nonempty cell of size r weight u_r . Define

$$\mathcal{F}_{\mathbf{u}}(q, t) = \sum_{n \geq 1} \sum_{P \in \text{PM}_n} q^{\text{inv}(P)} t^n \prod_{C \neq \emptyset} u_{|C|}.$$

For one column on $[k]$, its multiplicities give

$$R_k^{\mathbf{u}}(q, t) = \sum_{a_1, \dots, a_k \geq 0} \begin{bmatrix} a_1 + \dots + a_k \\ a_1, \dots, a_k \end{bmatrix}_q t^{a_1 + \dots + a_k} \prod_{i=1}^k u_{a_i}.$$

If

$$E_{\mathbf{u}}(z) = \sum_{a \geq 0} \frac{u_a z^a}{(q; q)_a},$$

then the word-content factorization and Lemma 2 give

$$\mathcal{B}_q R_k^{\mathbf{u}} = E_{\mathbf{u}}(z)^k, \quad R_k^{\mathbf{u}} = (q; q)_\infty \sum_{r \geq 0} \frac{q^r}{(q; q)_r} E_{\mathbf{u}}(q^r t)^k.$$

Theorem 4 (Cell cardinalities). *The complete cell-size enumerator is*

$$\mathcal{F}_u(q, t) = \sum_{d \geq 1} \prod_{k=1}^d \left(1 - (R_k^u(q, t))^{-1}\right).$$

Proof. The multiplicity of letter i in a column word is the size of the corresponding cell, so the displayed column series gives exactly the required cell weights. Forbidding a row simply removes a possible letter; absent cells have weight $u_0 = 1$. The same inclusion–exclusion and path summation therefore apply verbatim with R_k replaced by R_k^u . $\square$

Taking $u_r = 1$ for all $r \geq 1$ recovers Theorem 1. Taking $u_r = y$ for every positive r records the number of occupied cells; the full refinement retains the strictly finer complete size profile.

4 Boundary values and scope

At $q = 0$, only weakly increasing words remain, so

$$R_k(0, t) = (1 - t)^{-k}, \quad \mathcal{F}(0, t) = \sum_{d \geq 1} \prod_{k=1}^d (1 - (1 - t)^k).$$

At $q = 1$, every length- m word has weight one, so

$$R_k(1, t) = (1 - kt)^{-1}, \quad \mathcal{F}(1, t) = \sum_{d \geq 1} d! t^d.$$

The latter follows from the word definition; the q -adic product in the hypergeometric expression is not evaluated by direct substitution at $q = 1$.

The original sum-product solution is recorded in the released proof [1]. The normalized q -Borel transform, Euler kernel, Rogers–Szegő factorization, and basic hypergeometric notation are classical; see Gasper–Rahman [5] and Vinroot [3]. The contribution here is their direct use in the partition-matrix sum-product and the resulting cell-cardinality extension. No literature-wide novelty claim is made.

References

- [1] [Released partition-matrix sum-product proof](#), fixed revision [f6179e8](#).
- [2] S. Chern and S. Fu, *Signed counting of partition matrices*, Question 5.1, [arXiv:2508.21318v2](#).
- [3] C. R. Vinroot, *Multivariate Rogers–Szegő polynomials and flags in finite vector spaces*, Electron. J. Combin. 19 (2012), no. 3, P5; [arXiv:1011.0984](#).
- [4] S. Janson, *Generalized Galois numbers, inversions, lattice paths, Ferrers diagrams and limit theorems*, Electron. J. Combin. 19 (2012), no. 3, P34.
- [5] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, second edition, Cambridge University Press, 2004.